Supplementary Table 3: Investigations and management strategies

	Overall AIE	Definite AIE combined	Definite AIE									Possible AIE	Bickestaff's encephalitis
			NMDAR*	LGI1	CASPR2	GAD65	GFAP	Amphiphysin	Myelin	Yo	Non-specific		
EEG	Abnormal (36/43, 83.7%) Asymmetry (12/43, 27.9%) Slowing (27/43, 62.8%) IEDs (5/43, 11.6 %) Temporal slowing (9/43, 20.9%) Temporal IED (1/43. 2.3%) LPD (6/43, 13.9%), Frontal slowing (8/43, 18.6%)	Abnormal 11/14 (78.6) Slowing 8/14 (57.1) Temporal IED 1/14 (7.1)	Abnorm (6/7, 85.7%) Slowing (3/7, 42.9%) Temporal IED (1/7, 14.3%)	Fronto-temporal slowing	Diffuse slowing	Slowing (2/3, 66.7)	N.A.	Normal	N.A.	N.A.	Temporal slowing	Abnormal 25/29 (86.2%) Asymmetry 8/29 (27.6%) Slowing 17/29 (58.6%) Temporal slowing 5/29 (17.2%) Frontal slowing (4/29, 13.8%) IEDs (3/29, 10.3%), LPDs (5/29, 17.2%), NCSE (1/29, 3.4%) Triphasics (1/29, 3.4%)	Diffuse slowing in 1 case
MRI brain	Abnormal 37/64 (57.8) Supratentorial pathology 35/64 (54.7) Infratentorial 9/64 (14.1) Both 5/64 (7.8) Temporal lobe involvement 17/64 (26.6) Brainstem signal changes 8/64 (12.5) Pontine involvement 7/64 (10.9) Cerebellar 3/64 (4.7) Splenial involvement 3/64 (4.7) Enhancement 2/64 (2.3) Bleed 2/64 (3.1) Diffusion restriction 2/64 (3.1)	Abnormal 6/18 (33.3) Temporal involvement 4/18 (22.2)	Abnormal 4/11, Temporal involvement 4/11, Bilateral in 3/11 cases	Normal	Normal	Brainstem, cerebellar involvement in 1 case Bilateral cingulate and anterior temporal signal changes in 1 case 1 case normal	Normal	Normal	Fornix involvement in 1 case Normal in another case	Normal	Normal	Abnormal 31/47 (66.0), Supratentorial involvement 29/47 (61.7) Temporal signal changes 13/47 (27.7) Brainstem signal changes 7/47 (14.9)	Cerebellar involvement in 1 case, normal in 2 cases
PET	Done in 11 cases	Basal ganglia	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.	Basal ganglia	N.A.	N.A.	Hypometabolism	N.A.

	(15.5%) Hypometabolism in bilateral prefrontal cortex 3/11 (27.3) Hypermetabolism BG 4/11 (36.4) Hypermetabolsim limbic regions 1/11 (9.1) Cerebellar hypermetabolism 5/11 (45.5) Vermian involvement 4/11 (36.4)	hypermetabolism in a single case with anti-myelin.							hypermetabolism (1 case) Normal in another case			in bilateral prefrontal cortex 3/10 (30.0) Hypermetabolism BG 4/10 (40.0) Hypermetabolsim limbic regions 1/10 (10.0) Cerebellar hypermetabolism 5/10 (50.0) Vermian involvement 4/10 (40.0)	
Treatment													
At least one form of immunotherapy	47/57[£] (82.5)	19/19 (100)[*]	11/11[*] (100)	1 (100)	2 (100)	3[*] (100)	1(100)	1 (100)	NA	NA	1/1(100)	29/38 (76.3)	2/2 (100)
Parenteral steroids[&]	39/57 (68.4)	17/19 (89.5)	8/11 (72.7)	1 (100)	2(100)	3 (100)	1	1	NA	NA	1 (100)	23/38 (60.5)	1
IVIg	24/57 (42.1)	13/19 (68.4)	8/11 (72.7)	0	1 (50)	2 (66.7)	0	0	NA	NA	1 (100)	12/38 (30.8)	1
IVIg 2nd course	3/57 (5.3)	3/19 (15.8s)	0	0	0	2 (66.7)	0	0	NA	NA	1 (100)	0	0
Plasma exchange	8/57 (14.3)	5/19 (26.3)	2/11 (18.2)	0	1 (50)	1 (50)	0	0	NA	NA	1 (100)	3/38 (7.7)	0
Combination[$]	21/57 (36.8)	12/19 (63.2)	7/11 (63.6)	0	2 (100)	2 (100)	0	0	NA	NA	1 (100)	10/38 (25.6)	0
No immunotherapy	9/ 57 (15.8)	--	--	--	--	--	--	--	NA	NA	0	9/38 (23.1)	0
Clinical outcomes													
Immunotherapy responders	42/48 (87.5)	18/19 (94.7)	10/11 (90.9)	1 (100)	2 (100)	3 (100)	1 (100)	1 (100)	N.A.	N.A.	1 (100)	24/29 (86.2)	2/2 (100)
Latency to therapeutic response in Median (IQR) days	8 (5-19) (36/71 cases)	11 (7-30) (15/23 cases)	NA	NA	NA	NA	NA	NA	NA	NA	NA	7 (5-11) (21/48 cases)	NA
Sequelae[#]	22/60 (36.7)	6/15 (40.0)	4/11 (36.4)	0	0	2/3 (66.7)	0	1 (100)	N.A.	N.A.	0	16/45 (35.6)	1/3(33.3)
Expired	4 (5.6)	0	0	0	0	0	0	0	0	0	0	4 (8.3)	0